

Flour Mill SCADA - Production Grade Rules

1. Overview

This document outlines the strict machine efficiency thresholds required for different flour grades. The central SCADA system tracks subpart efficiency across all 10 major milling machines.

2. Ultra-Fine Cake Flour (Premium Grade)

Requirement: Requires absolute precision in particle size distribution and zero bran contamination.

Machine Thresholds:

- RM-301 and RM-302 (Roller Mills): Corrugated and Smooth Grinding Rolls MUST operate at 100% efficiency. Anything lower will result in off-spec coarse particles.
- PUR-400 (Purifier): Fine and Coarse Sieve Mesh MUST operate at 100% efficiency to guarantee zero ash/bran content.
- PS-500 (Plansifter): Sieve stacks must be >98% efficiency.

3. Standard Bread Flour (Commercial Grade)

Requirement: Standard industrial baking grade. Tolerates minor variance.

Machine Thresholds:

- Roller Mills (RM-301, RM-302): Grinding rolls can operate down to 90% efficiency. Roll gaps are more forgiving.
- Purifier (PUR-400): Sieve meshes can operate down to 85% efficiency without compromising commercial standards.
- Pneumatic Blower (PB-700): Inlet filters must remain above 80% to maintain adequate conveying velocity.

4. Animal Feed Grade (Lowest Grade)

Requirement: High volume, low precision.

Machine Thresholds:

- All Roller Mill and Purifier subparts can operate down to 70% efficiency.
- Bran Finisher (BF-600) must operate above 80%.